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Phase equilibria in the Fe–Co binary system

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Abstract

α (A2)/γ (A1) phase equilibria of the Fe–Co system between 400 and 800°C were determined by means of lattice parameter measurement using thin film specimens. Bulk specimens were also analyzed to compare the extent of attainment to the equilibrium. The thin film technique was found to be greatly advantageous for obtaining the phase equilibria at lower temperatures where solid-state reactions are too slow to reach the equilibrium state in the conventional methods using bulk specimens. It was confirmed that the α + γ two-phase region extends below the temperature at which the α (A2)/α′ (B2) transus meets the α/α + γ boundary. Thermodynamic analysis was also conducted by taking the magnetic and chemical ordering contributions of the B2 structure into account, findings of which confirmed the extension of the α + γ two-phase region below the α/α′ ordering temperature. © 2002 Acta Materialia Inc. Published by Elsevier Science Ltd. All rights reserved.

Keywords: Phase equilibria; Order–disorder phenomena; Thermodynamics; Thin films; Microstructure

1. Introduction

The phase diagram of the Fe–Co system has been extensively investigated because of its practical importance in structural and functional alloys, especially magnetic materials, and particular interest has been generated because of the apparent contributions of magnetic and chemical ordering to the phase equilibria. Fig. 1 shows a compiled phase diagram of the Fe–Co system [1]. It has been suggested that the peculiar shape of the α (A2)/γ (A1) phase boundary in the Fe–Co system, i.e., Co raises both the A3 and A4 temperatures, is mainly due to the magnetic contribution. Kaufman and Nesor [2] have reported pioneering work on the phase diagram of the Fe–Co system using an empirical equation to describe the thermodynamics parameters. In their assessment, the magnetic contribution to the Gibbs energy was not taken into account, which resulted in a complex temperature dependence of the lattice stability and interaction parameters. Inden [3] and Hillert and Jarl [4] proposed a thermodynamic description of the mag-

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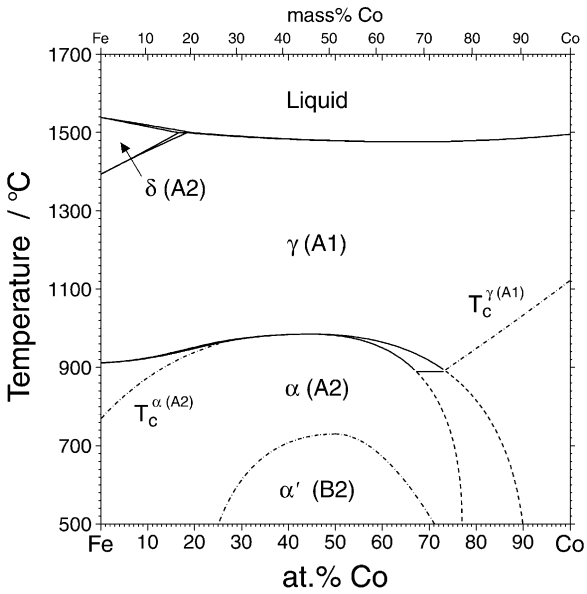

Fig. 1. Phase diagram of the Fe–Co binary system [1].

netic contribution and Fernandez Guillermet applied it for the thermodynamic assessment of the Fe–Co system [5]. In his assessment, however, the α (A2)/α′ (B2) order–disorder transition was not taken into account. The α/α′ boundary was precisely examined by Ellis and Greiner [6], Asano et al. [7], Oyedale and Collins [8] and Mills et al. [9], and thermodynamic calculations using the cluster variation method (CVM) [10] and coupled description [11] of CVM and CALPHAD were performed by Colinet et al., although they did not present the effect of the ordering on the low-temperature α (A2)/γ (A1) equilibrium. Their calculation shows that the α (A2)/α′ (B2) transus under a fictitious non-magnetic condition deviate to a lower temperature than that of the magnetic system [10], which indicates the presence of an interaction between the magnetic and chemical ordering suggested in our previous paper [12].

Experimental determination of the phase equilibria at low temperatures is very difficult because of the low diffusivity where the conventional method using bulk specimens cannot be applied. The thin-film method, however, might overcome such difficulty due to the enhanced role of surface and grain boundary diffusion. The thin-film method for examining the ternary phase diagram was first reported by Kennedy et al. [13]. They demonstrated that the equilibrium state could be reached rapidly by annealing thin-film specimens and succeeded in determining isothermal sections of the Fe–Ni–Cr system. It is expected, therefore, that this technique enables attainment of low-temperature equilibria between the α′ (B2) and γ (A1) phases. The purpose of the present study was to examine the effect of the A2/B2 ordering reaction on the b.c.c. (α and α′)/f.c.c. (γ) equilibria by experiment and thermodynamic calculations.

2. Experimental procedure

Thin films with a thickness of about 0.4–1.0 μm were prepared on a rock-salt substrate by a conventional DC magnetron sputtering method, the composition of the films being controlled at Co–22 mass% Fe so as to have the α + γ two-phase microstructure by equilibration between 400 and 800°C. After the evacuation to a pressure of 1.1×10^{-4} Pa, Ar gas was conducted into the chamber to 4.0–5.3 Pa and the deposition was carried out with an applied voltage of 610 V and a discharge current of 0.15 A. The temperature of the substrate was kept at about 350°C during the deposition and the deposition rate was about 4 nm/min. After dissolving the rock-salt substrate, the specimens were encapsulated in quartz tubes and annealed at temperatures between 400 and 800°C.

Bulk specimens of Co–(9–37) mass% Fe alloys were also prepared by induction melting and the calibration curves of the alloy composition against the lattice parameters were determined. The bulk sample of the Co–22 mass% Fe alloy, whose composition corresponds to that of the thin films, was also prepared for equilibration to compare the evolution to the equilibrium.

The lattice parameters of the b.c.c. (α and α′) and f.c.c. (γ) phases in both the thin-film and the bulk specimens were measured by the back-reflection focusing method with Co K_{α_1} radiation. Although the lattice spacing of film varies with its thickness due to surface stress in very thin film, this effect was found to be negligible for the film prepared in the present study. The crystallographic structures were determined by an X-ray dif-

fractometer and by electron diffraction. The microstructure of the thin-film specimens was examined by transmission electron microscopy (TEM), where the specimens were prepared by the ion beam thinning method.

3. Experimental results

3.1. Lattice parameters of b.c.c. and f.c.c. phases

Fig. 2 shows the lattice parameters of the f.c.c.-γ (A1) phase, a_α, in the bulk specimens annealed at 1000°C. The lattice parameters can be seen to increase linearly with increasing Fe content, which is in accordance with the previous work [6]. The relationship between the lattice parameter and the composition was determined to be

$$a_\gamma = 0.35438 + 0.00010233\times[\text{at.\% Fe}]/\text{nm}. \quad (1)$$

The lattice parameters of the b.c.c. phase, on the other hand, were reported to differ depending on the heat treatment conditions, which is considered to be caused by the α (A2)/α′ (B2) order–disorder transition. In the present study, therefore, two kinds of condition were conducted: (i) as quenched after solution annealing at 800°C and (ii) followed by heat treatment at 400°C for 240 h below the order–disorder transition temperature. The results are shown in Fig. 3 together with the data of Asano et al. [7]. It is confirmed that the lattice parameter of the ordered structure, $a_{\alpha'}$, is larger than that of disordered one, a_α, which corresponds to the characteristic features of the Fe–Co ordering system. The following relation was obtained:

$$a_\alpha = 0.28236 + 0.000064514\times[\text{at.\% Fe}]/\text{nm}, \quad (2)$$

$$a_{\alpha'} = 0.28250 + 0.000064231\times[\text{at.\% Fe}]/\text{nm}. \quad (3)$$

3.2. Equilibration process in bulk and thin-film specimens

Thin-film and bulk specimens of Co–22 mass% Fe alloy were annealed at temperatures between 400 and 700°C for various durations and their microstructures were analyzed to compare the equilibrium conditions. The crystal structure of the bulk specimens pretreated at 1000°C was f.c.c., while that of thin-film specimens sputtered at about 350°C was b.c.c. as shown in Fig. 4. As-sputtered specimens had a very fine microstructure of 0.05 μm in grain radius approximately as shown in the TEM micrograph of Fig. 5, which resulted in the

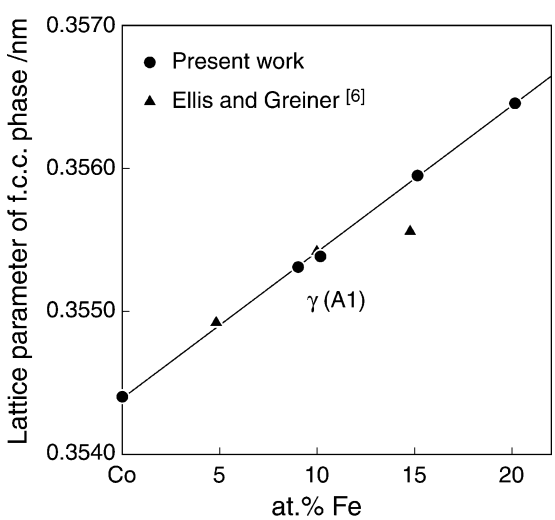

Fig. 2. Lattice parameter of the γ (A1) phase.

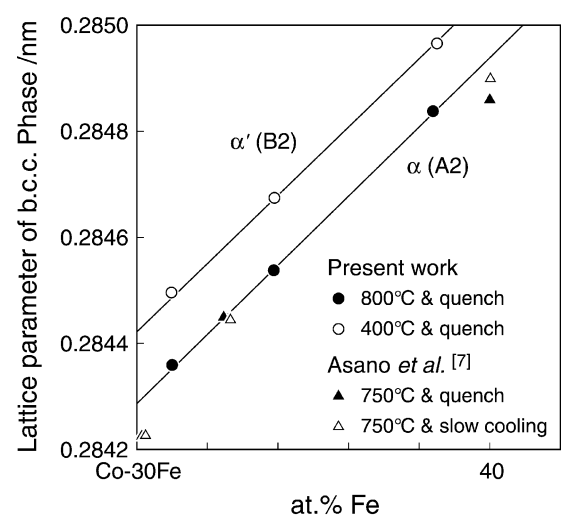

Fig. 3. Lattice parameter of the α (A2) and α′ (B2) phases.

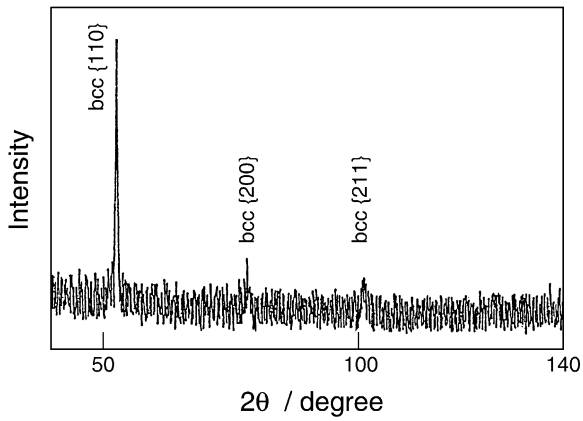

Fig. 4. X-ray diffraction pattern for the as-sputtered thin film of Co–22 at.% Fe alloy.

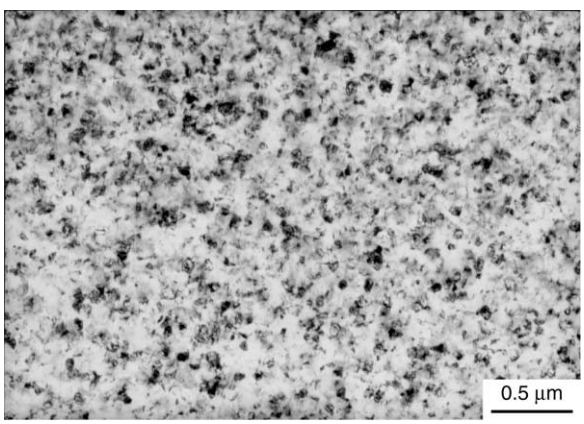

Fig. 5. TEM of as-sputtered thin film of Co–22 at.% Fe alloy.

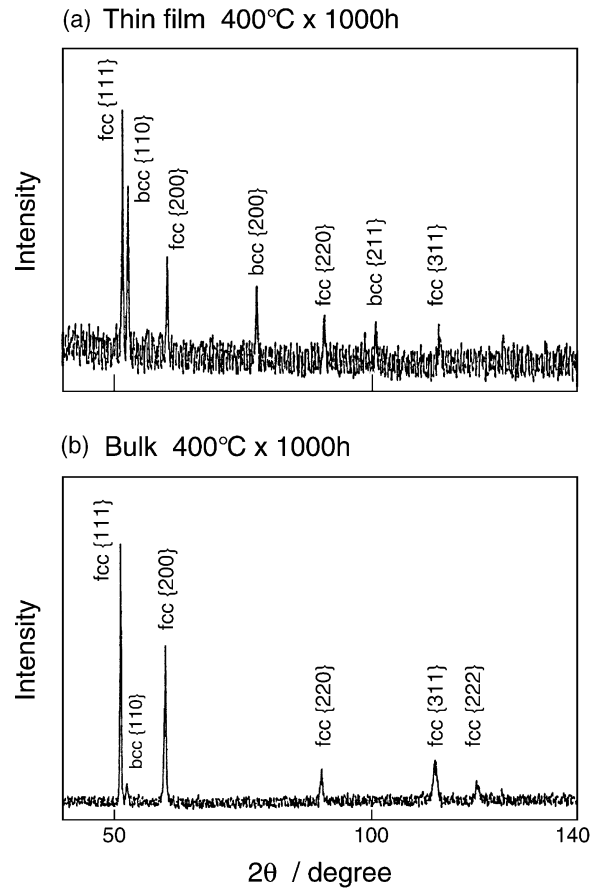

Fig. 6. X-ray diffraction patterns of thin film and bulk specimens of Co–22 at.% Fe alloy equilibrated at 400°C for 1000 h.

enhancement of the diffusion of elements in thin-film specimens. According to the results of X-ray diffraction of the thin-film and bulk specimens annealed at 700°C for 30 h, no remarkable difference was observed in the phase constitution of the $\alpha + \gamma$ two-phase structure. On the other hand, the result for the specimens annealed at 400°C for 1000 h shown in Fig. 6 reveals that the duration of the heat treatment of bulk specimens was insufficient to attain the $\alpha + \gamma$ equilibrium. Fig. 7 presents the TEM micrograph of a thin-film specimen annealed at 400°C for 2200 h showing an $\alpha + \gamma$ micro-duplex structure with a grain size of 0.2 μm. The composition of the α and the γ phases in all the specimens was determined using the cali-

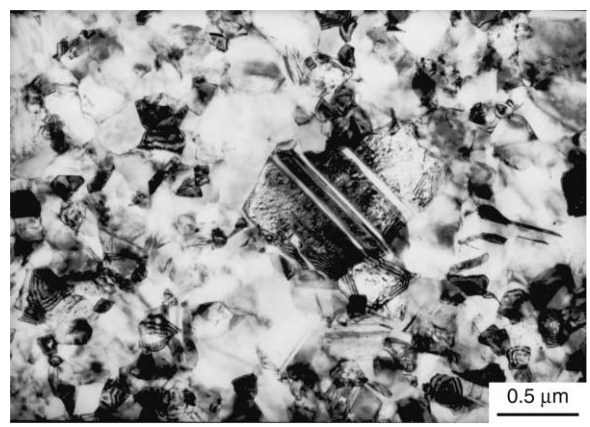

Fig. 7. Transmission electron micrograph of the thin film of Co–22 at.% Fe alloy annealed at 400°C for 2200 h.

bration curves described in Eqs. (1) and (2), except for the b.c.c. phase obtained at 400 and 450°C below the A2/B2 transition temperature, to which Eq. (3) for the ordered α′ phase was applied. Fig. 8(a) and (b) show the temporary compositions of each phase during heat treatment at 700 and 400°C, respectively. Equilibration of the thin-film specimens was attained within 1 h at 700°C, while the 30-h annealing was insufficient for the bulk specimens. The evolution to equilibrium at 400°C exhibits a remarkable difference between thin-film and bulk specimens. The composition of the γ phase in the bulk specimen hardly changed during heat treatment for 2200 h, which indicates that this heating time was insufficient to generate decomposition to the α + γ phases. In the case of the thin-film specimens, on the other hand, the compositions are gradually changed toward the equilibrium after the decomposition, but the 1000-h annealing is insufficient to attain the equilibrium. These results suggest that the lower limit of the annealing temperature in determining the phase boundary by the thin-film method is 400°C (≈0.4 T_m). The equilibrium compositions between α (A2), α′ (B2) and γ (A1) phases are listed in Table 1 and are shown in Fig. 9, in comparison with the previous data [6] and the calculated diagram [5]. The results of the energy dispersive X-ray (EDX) analysis of the two-phase bulk specimens are also included.

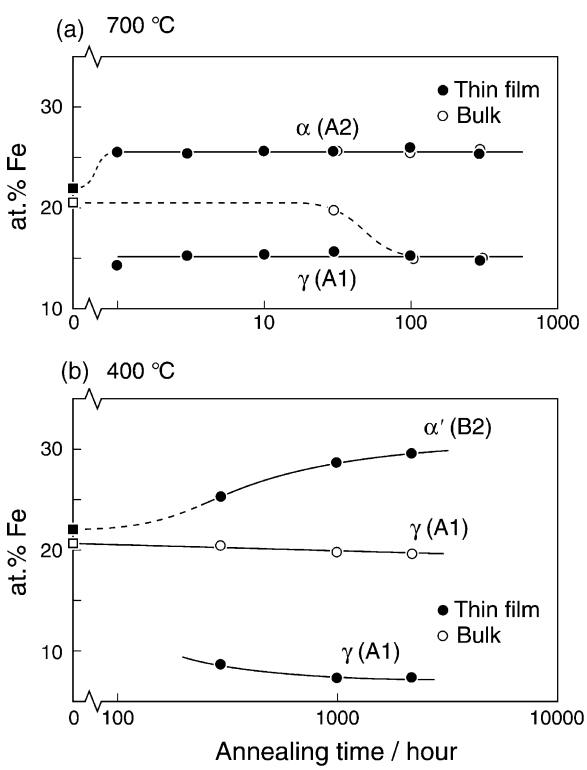

Fig. 8. Evolution to the equilibrium state with increasing annealing time.

4. Thermodynamic assessment

The Gibbs energies of the stable and metastable structures for the pure elements are taken from the SGTE database compiled by Dinsdale [14]. In the Fe–Co system, four disordered solution phases, the liquid, α (A2), γ (A1) and ε (A3), and an ordered b.c.c. phase, α′ (B2), are considered to constitute the equilibrium phases. The Gibbs energy descriptions of these five phases are given below and evaluated thermodynamic parameters are listed in Tables 2 and 3.

4.1. Disordered solution phases

The liquid, α (A2), γ (A1) and ε (A3) phases are modeled as substitutional solutions and the Gibbs energy without the magnetic contribution is described by the sub-regular solution approximation

$$G_m^\phi = {}^\circ G_{Co}^\phi \cdot x_{Co} + {}^\circ G_{Fe}^\phi \cdot x_{Fe} + RT(x_{Co} \ln x_{Co} + x_{Fe} \ln x_{Fe}) + x_{Co} x_{Fe} \cdot L_{CoFe}^\phi, \quad (4)$$

where ${}^\circ G_{Co}^\phi$ and ${}^\circ G_{Fe}^\phi$ are the Gibbs energies of pure Co and pure Fe, respectively, in the corresponding structure of the disordered solution ϕ relative to their reference states. x_{Co} and x_{Fe} are the mole fractions of Co and Fe, respectively. L_{CoFe}^ϕ represents the interaction parameter which has a composition dependence in the form of the Redlich–Kister (RK) polynomial described by the following equation:

$$L_{CoFe}^\phi = \sum_n (x_{Co} - x_{Fe})^n \cdot {}^n L_{CoFe}^\phi. \quad (5)$$

Coefficients, ${}^n L_{CoFe}^\phi$, are evaluated on the basis of

Table 1
Experimental data of the equilibrium compositions between the b.c.c. and f.c.c. phases

Temperature (°C)	b.c.c./at.% Fe		f.c.c./at.% Fe	
	Thin film	Bulk	Thin film	Bulk
800	27.6	26.4[a]	20.5	19.6[a]
700	25.5	25.5	14.9	14.9
700		26.0[a]		15.1[a]
600	25.3		10.9	
550	26.1		10.9	
500	26.3		9.6	
450	29.1[b]		9.2	
400	29.6[b]		7.3	

[a] Determined by EDX analysis.
[b] With reference to the lattice parameter of α′ (B2) phase.

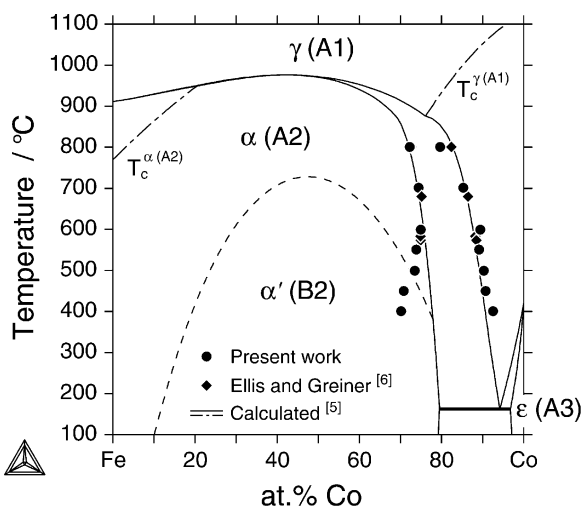

Fig. 9. Summary of the experimental results on the phase equilibria between γ (A1), α (A2) and α′ (B2) phases of the Fe–Co binary system in comparison with the calculated diagram [5].

the experimental data on phase boundaries and thermochemical properties.

4.2. Ordered phases

The Gibbs energy of the ordered α′ (B2) phase is described by the two-sublattice split compound energy formalism (s-CEF) which was proposed by Ansara et al. [15]:

$$G_m^{\Phi} = G_m^{\phi}(x_i) + \Delta G_m^{ord}(y_i^s), \quad (6)$$

Table 2
Thermodynamic and magnetic interaction parameters of the Co–Fe system evaluated in this study. Lattice stabilities and the magnetic properties of pure elements are taken from [14]

Thermodynamic and magnetic interaction parameters	
Liquid	
L (LIQUID,CO,FE;0)	−9753.82
L (LIQUID,CO,FE;2)	+2757.96
α-(A2)	
L (A2—BCC,CO,FE;0)	$-26{,}222.7+125.000T-15.5020T \ln T+0.25\Delta G_{CoFe}^{SRO}$
L (A2—BCC,CO,FE;2)	$+2686.79-0.25\Delta G_{CoFe}^{SRO}$
TC (A2—BCC,CO,FE;0)	+590 [5]
BMAGN (A2—BCC,CO,FE;0)	+1.406
BMAGN (A2—BCC,CO,FE;1)	−0.6617
γ-(A1)	
L (A1—FCC,CO,FE;0)	−8968.75
L (A1—FCC,CO,FE;2)	+3528.80
TC (A1—FCC,CO,FE;0)	+283 [5]
TC (A1—FCC,CO,FE;1)	+879 [5]
BMAGN (A1—FCC,CO,FE;0)	+8.407
BMAGN (A1—FCC,CO,FE;1)	−3.644
ε-(A3)	
L (A3—HCP,CO,FE;0)	+5000
TC (A3—HCP,CO,FE;0)	−253 [5]
TC (A3—HCP,CO,FE;1)	+1494 [5]
BMAGN (A3—HCP,CO,FE;0)	+5.41 [5]
BMAGN (A3—HCP,CO,FE;1)	−0.24 [5]

Table 3
Thermodynamic and magnetic parameters of the B2 ordering contribution of the Co–Fe system evaluated in this study

Parameters of B2 ordering on Gibbs energy and magnetic properties	
α′-FeCo (B2)	
G (B2—BCC,CO:CO;0), G (B2—BCC,FE:FE;0)	+0.0
G (B2—BCC,CO:FE;0), G (B2—BCC,FE:CO;0)	−1621.23
L (B2—BCC,CO:CO,FE;0), L (B2—BCC,CO,FE:CO;0)	+1744.89
L (B2—BCC,CO:CO,FE;1), L (B2—BCC,CO,FE:CO;1)	−123.653
L (B2—BCC,FE:CO,FE;0), L (B2—BCC,CO,FE:FE;0)	+1497.58
L (B2—BCC,FE:CO,FE;1), L (B2—BCC,CO,FE:FE;1)	−123.653
L (B2—BCC,CO,FE:CO,FE;0) ($= \Delta G^{\mathrm{SRO}}_{\mathrm{CoFe}}$)	-2.5290×10^{6} T^{-1}
TC (B2—BCC,CO:CO;0), TC (B2—BCC,FE:FE;0)	+0.0
TC (B2—BCC,CO:FE;0), TC (B2—BCC, FE:CO;0)	+370
TC (B2—BCC,CO:CO,FE;0), TC (B2—BCC,CO,FE:CO;0)	−370
TC (B2—BCC,FE:CO,FE;0), TC (B2—BCC,CO,FE:FE;0)	−370
BMAGN (B2—BCC,CO:CO;0), BMAGN (B2—BCC,FE:FE;0)	+0.0
BMAGN (B2—BCC,CO:FE;0), BMAGN (B2—BCC, FE:CO;0)	+0.1
BMAGN (B2—BCC,CO:CO,FE;0), BMAGN (B2—BCC,CO,FE:CO;0)	−0.1
BMAGN (B2—BCC,FE:CO,FE;0), BMAGN (B2—BCC,CO,FE:FE;0)	−0.1

where $G^{\phi}_{\mathrm{m}}(x_i)$ is the Gibbs energy of the disordered solution ϕ of the primitive structure corresponding to the ordered phase Φ. $\Delta G^{\mathrm{ord}}_{\mathrm{m}}(y^s_i)$ is the ordering energy extracted from the Gibbs energy described by the compound energy formalism (CEF), $G^{\mathrm{CEF}}_{\mathrm{m}}(y^s_i)$, represented by the following equation [15]:

$$\Delta G^{\mathrm{ord}}_{\mathrm{m}}(y^s_i) = G^{\mathrm{CEF}}_{\mathrm{m}}(y^s_i) - G^{\mathrm{CEF}}_{\mathrm{m}}(y^s_i = x_i), \tag{7}$$

where $G^{\mathrm{CEF}}_{\mathrm{m}}(y^s_i = x_i)$ is the order-independent contribution of $G^{\mathrm{CEF}}_{\mathrm{m}}(y^s_i)$. y^s_i represents the site fraction of element i on the sth sublattice. The Gibbs energy of the ordered phase is expressed as the sum of two contributions, one ordering term and one disordering term, the evaluated Gibbs energy of the disordered phases described by Eq. (1) being available for the latter. In this unified description, thermodynamic properties and phase equilibria of the ordered and disordered phases can be evaluated separately.

4.3. Magnetic contribution

The magnetic contribution to the Gibbs energy, $G^{\mathrm{mag}}_{\mathrm{m}}$, was introduced into the α (A2), γ (A1) and ε (A3) disordered solution phases. Its description was originally suggested by Inden [3] and the description modified by Hillert and Jarl [4] is applied to the calculation as follows:

$$G^{\mathrm{mag}}_{\mathrm{m}} = RT\ln(\beta + 1)f(\tau), \tag{8}$$

$$f(\tau) = 1 - \frac{1}{A}\left\{\frac{79\tau^{-1}}{140p} + \frac{474}{497}\left(\frac{1}{p} - 1\right)\left(\frac{\tau^3}{6} + \frac{\tau^9}{135} + \frac{\tau^{15}}{600}\right)\right\}, \text{ for } \tau = \frac{T}{T_{\mathrm{c}}} \le 1, \tag{9}$$

$$f(\tau) = -\frac{1}{A}\left(\frac{\tau^{-5}}{10} + \frac{\tau^{-15}}{315} + \frac{\tau^{-25}}{1500}\right), \text{ for } \tau = \frac{T}{T_{\mathrm{c}}} \ge 1 \tag{10}$$

$$A = \frac{518}{1125} + \frac{11692}{15975}\left(\frac{1}{p} - 1\right),$$

where T_{c} is the Curie temperature and β is the mean atomic moment expressed in Bohr magneton unit μ_{B}. Both values are functions of alloy composition as follows:

$$T_{\mathrm{c}} = {}^{\circ}T^{\mathrm{Co}}_{\mathrm{c}}\cdot x_{\mathrm{Co}} + {}^{\circ}T^{\mathrm{Fe}}_{\mathrm{c}}\cdot x_{\mathrm{Fe}} + x_{\mathrm{Co}}\cdot x_{\mathrm{Fe}}\sum_n {}^{n}T^{\mathrm{CoFe}}_{\mathrm{c}}(x_{\mathrm{Co}} - x_{\mathrm{Fe}})^n, \tag{11}$$

$$\beta = {}^{\circ}\beta_{\mathrm{Co}}\cdot x_{\mathrm{Co}} + {}^{\circ}\beta_{\mathrm{Fe}}\cdot x_{\mathrm{Fe}} + x_{\mathrm{Co}}\cdot x_{\mathrm{Fe}}\sum_n {}^{n}\beta_{\mathrm{CoFe}}(x_{\mathrm{Co}} - x_{\mathrm{Fe}})^n. \tag{12}$$

p represents the ratio of the magnetic enthalpy due to short range ordering to the total amount of magnetic enthalpy where the values $p = 0.4$ for b.c.c. and $p = 0.28$ for f.c.c. and h.c.p. alloys were suggested by Inden [3].

The magnetic properties, T_c and β, of pure Co and Fe with the stable and some metastable crystal structures were evaluated by direct and indirect measurements as shown in Table 4 [16–25]. When the magnetic moment of pure Co, $^{\circ}\beta_{Co} = 1.739$ is applied to Eq. (12), a calculated magnetic specific heat capacity of f.c.c.–Co deviates from empirical measurements. Therefore, an effective value for the calculation, $^{\circ}\beta^{eff}_{Co}$, was derived on the basis of a thermochemical assessment and was assumed to hold also in the b.c.c. and h.c.p. metastable structures [26]. In the present assessment, the magnetic moment of b.c.c. and f.c.c. disordered solutions was re-evaluated to coincide with the empirical measurements [20,27,28], and then, effective functions were derived by replacing $^{\circ}\beta_{Co}$ with $^{\circ}\beta^{eff}_{Co}$ as shown in Fig. 10(a) and (b).

In the description of Eqs. (11) and (12), the effects of the atomic order–disorder on the magnetic properties were not taken into account. It is suggested, however, that both T_c and β depend on the degree of ordering [29,30]. The thermodynamic calculation of this effect has been discussed in detail in our previous paper [12]. In the Fe–Co system, Goldman and Smoluchowski [31,32] revealed the effect of the B2 ordering on the saturation magnetization and suggested the saturation moment of the ordered alloy should be about 4% higher than in the disordered alloy. Eguchi et al. [33] and Bardos [27] carried out the quantitative measurements of the deviation between ordered and disordered states, which confirms Goldman's suggestion. Furthermore, the results of the Bragg–Williams (B-W) [33] and the cluster variation method (CVM) [10] calculation on the α (A2)/α′ (B2) boundary under an assumption of a fictitious non-magnetic state deviates lower from that of the actual ferro-magnetic condition. This indicates that the interaction between magnetic and chemical ordering enhances the stability of the ordered α′ (B2) phase [12]. The effect of the interaction was introduced into the present calculation by considering $\Delta T_c^{B2/A2} = T_c^{B2} - T_c^{A2}$ and $\Delta\beta^{B2/A2} = \beta^{B2} - \beta^{A2}$ using the two-sublattice s-CEF in the same way as for the ordering contribution to the Gibbs energy shown in Section 4.2. The value of the deviation of the equiatomic FeCo compound, $\Delta\beta^{B2/A2} = 0.1\mu_B$, about 4% of the mean atomic moment of Fe–Co disordered alloys, was accepted, which results in the good agreement between empirical [27] and calculated magnetic moments of the ordered B2 phase as shown in Fig. 10(a). The value of $\Delta T_c^{B2/A2}$ was evaluated by means of the thermochemical optimization in the present work. The evaluate parameters of the magnetic properties are listed in Tables 2 and 3.

4.4. Approximation of the Gibbs energy of SRO

The ordering contribution introduced by the split compound energy formalism (s-CEF) contains only

Table 4
Assessed magnetic properties of pure cobalt and iron determined by means of direct and indirect measurement. Values put in parentheses were optimized for calculation [5,14]

		f.c.c. (A1)	b.c.c. (A2)	h.c.p. (A3)
Co	T_c (K)	1394 [16] (1396)	1450 [18] (1450)	1070 [20] (1396)
	β/μ_B	1.739 [17] (1.35)	1.80 [19] (1.35)	1.715 [21] (1.35)
Fe	T_c, T_N[a] (K)	67[a] [22,23] (67[a])	1043 [21] (1043)	100[a] [25] (0)
	β/μ_B	0.7 [24] (0.7)	2.218 [21] (2.22)	0.1 [25] (0)

[a] Néel temperature.

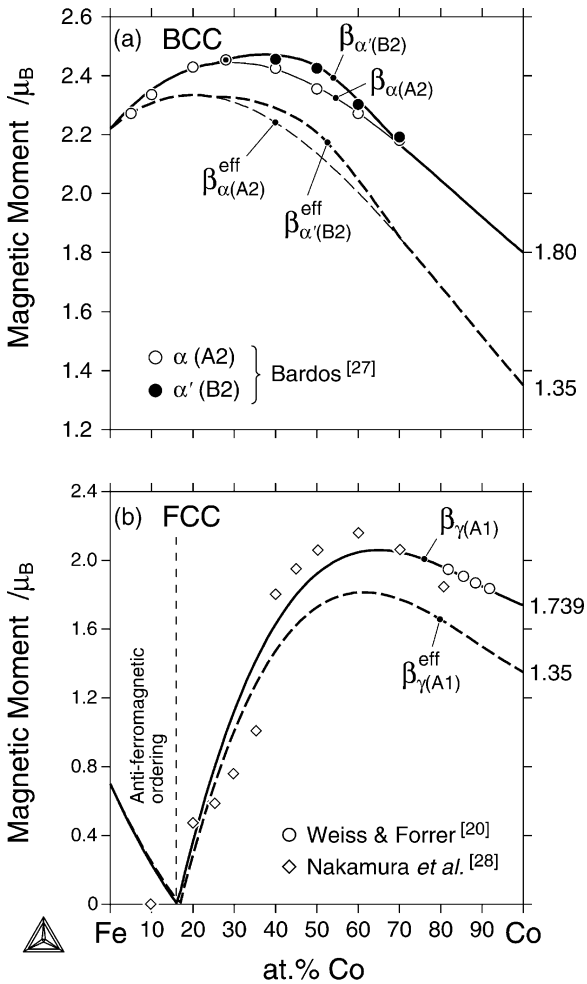

Fig. 10. Mean magnetic moment of (a) b.c.c. phases [26] and (b) f.c.c. phase [20,27] assessed in the present work (solid lines). Effective values for thermodynamic calculation are shown by dashed lines.

the long-range ordering (LRO) energy. Thus, the effect of the short-range ordering (SRO) is compensated for by the ordering contribution of the LRO description, which results in the deviation of the calculated order–disorder boundary from the practical one determined by experiments, as reported for the A2/B2 order–disorder boundary in the Ti–Al binary system [34]. The CVM describes both the LRO and the SRO contribution correctly and yields a satisfactory result in the calculation of the order–disorder system such as A1/$L1_0$/$L1_2$ equilibria in the Au–Cu binary system [35]. This method, however, entails some difficulties in the treatment of the magnetic contribution and the multi-component systems. Sundman et al. [36] proposed that the effect of the SRO due to pair probabilities can be approximated by introducing reciprocal parameters into the CALPHAD method and reported the thermodynamic assessment of the Au–Cu system. This approximation is based on the quasi-chemical formalism for the ionic liquid phase [37,38], and the effect of the SRO for the two-sublattice CEF is described as

$$\Delta G_{\mathrm{m}}^{\mathrm{SRO}} = -y_{\mathrm{A}}^{\mathrm{I}} y_{\mathrm{A}}^{\mathrm{II}} y_{\mathrm{B}}^{\mathrm{I}} y_{\mathrm{B}}^{\mathrm{II}} \times \frac{(\Delta^{\circ} G_{\mathrm{m}}^{\mathrm{AB}})^2}{zRT}, \tag{13}$$

where z represents the coordination number and R is the gas constant. The relation of $\Delta^{\circ}G_{\mathrm{m}}^{\mathrm{AB}}$ to the Gibbs energy of formation is represented by the following equation:

$$\Delta^{\circ} G_{\mathrm{m}}^{\mathrm{AB}} = {}^{\circ}G_{\mathrm{A:B}} + {}^{\circ}G_{\mathrm{B:A}} - {}^{\circ}G_{\mathrm{A:A}} - {}^{\circ}G_{\mathrm{B:B}}. \tag{14}$$

The reciprocal component of the Gibbs energy described by the two-sublattice s-CEF, $-(\Delta^{\circ}G_{\mathrm{m}}^{\mathrm{AB}})^2/zRT$ ($= \Delta G_{\mathrm{CoFe}}^{\mathrm{SRO}}$), should affect not only on the ordering contribution, $\Delta G_{\mathrm{m}}^{\mathrm{ord}}(y_i^{\varsigma})$, but also on the Gibbs energy of the disordered solution, $G_{\mathrm{m}}^{\phi}(x_i)$. The contribution of $\Delta G_{\mathrm{CoFe}}^{\mathrm{SRO}}$ was involved in the Gibbs energy as shown in Tables 2 and 3, and evaluated on the basis of a thermochemical assessment.

5. Discussion

5.1. Phase equilibria between the α (A2), α′ (B2) and γ (A1) phases

As shown in the present result on the phase equilibria between α (A2), α′ (B2) and γ (A1) phases in Fig. 9, the α (A2) + γ (A1) two-phase region extends with decreasing temperature. Below the intersection with the α (A2)/α′ (B2) boundary, the equilibrium turns into α′ (B2) + γ (A1) and further enlargement of the two-phase region is observed. Compared with the previous assessment of the phase diagram of the Fe–Co system by Fernandez Guillermet [5], the b.c.c./b.c.c. + γ (A1) boundary

significantly deviates at low temperatures, such deviation being considered to be caused by the ordering reaction in the b.c.c. phase.

5.2. Comparison of calculated and experimental data

Fig. 11 shows the calculated and experimentally determined thermodynamic properties related to (a) the mixing enthalpy of liquid alloys at 1600°C [39,40] and the formation enthalpies of (b) f.c.c. solution at 1200°C [41], (c) b.c.c. solution at 870°C [41] and (d) b.c.c. A2 and B2 solution at 502°C [42]. Results of the calculation are in good agreement with the experimental data. Fig. 12 shows a comparison of the calculated phase equilibria (a) between the liquid, δ (A2) and γ (A1) phases and (b) between the γ (A1), α (A2), α′ (B2) and ϵ (A3) phases together with the experimental data [6–9,39,43,44]. Agreement between calculation and experiment is quite satisfactory not only for the phase equilibria but also for the A2/B2 transition boundary. It should be noted that the calculated b.c.c. (α and α′) + f.c.c. (γ) two-phase region

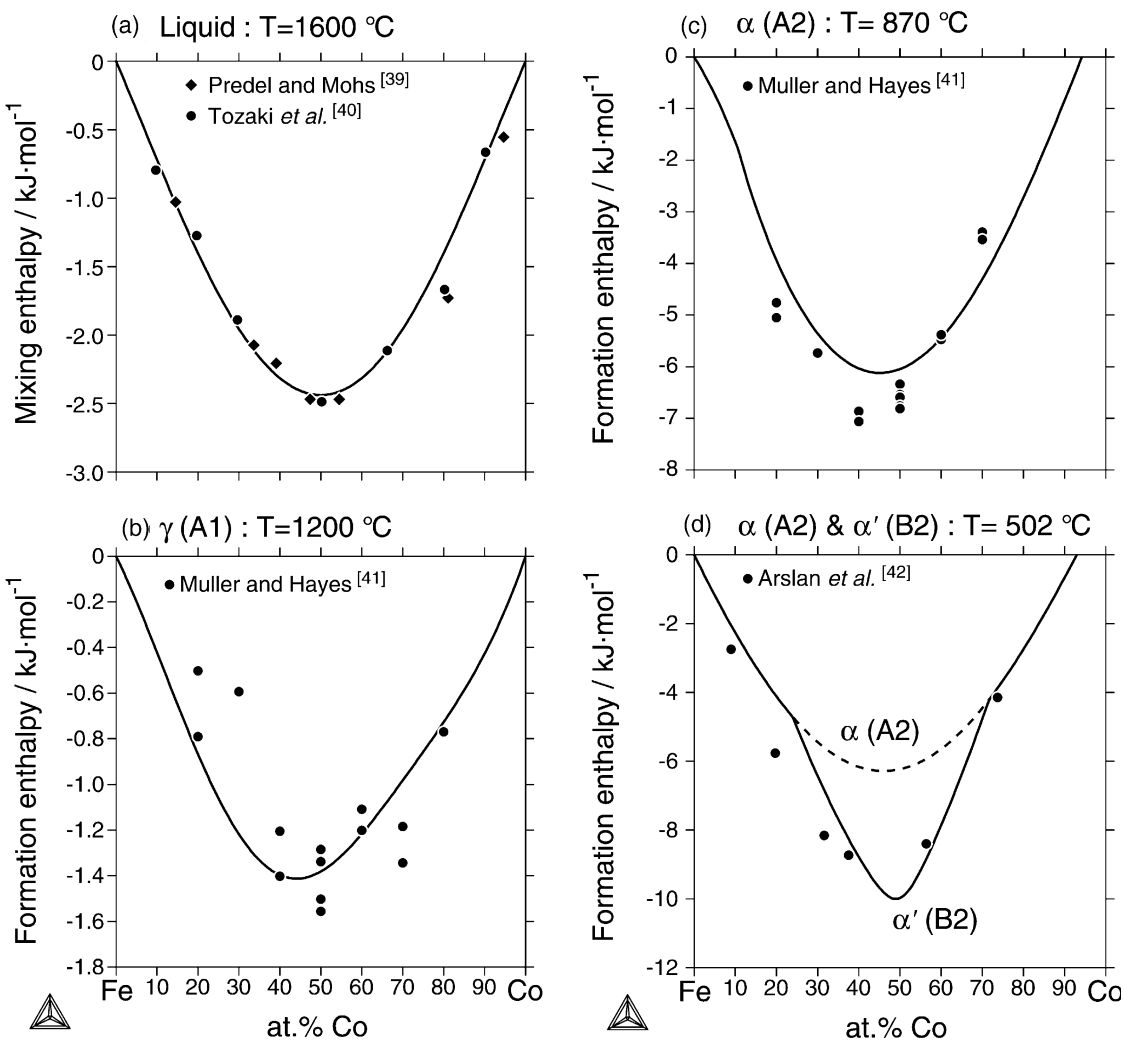

Fig. 11. Optimization results on the thermodynamic properties: (a) mixing enthalpy of liquid phase at 1600°C [38,39] and formation enthalpies of solid phases, (b) γ (A1) at 1200°C [40], (c) α (A2) at 870°C [40] and (d) α (A2) and α′ (B2) at 502°C [41].

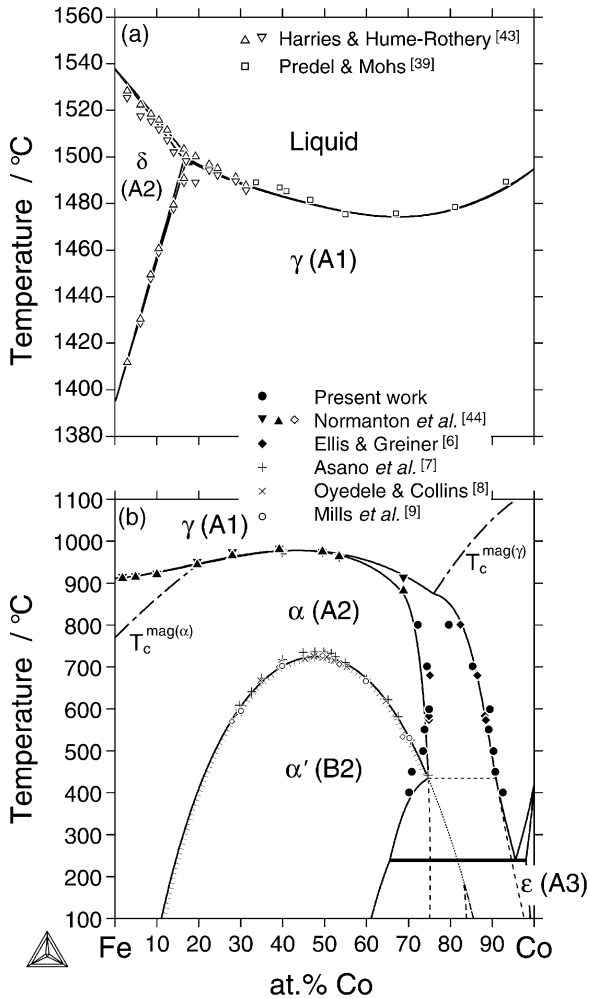

Fig. 12. Calculated phase diagrams of the Fe–Co binary system in comparison with the experimental data.

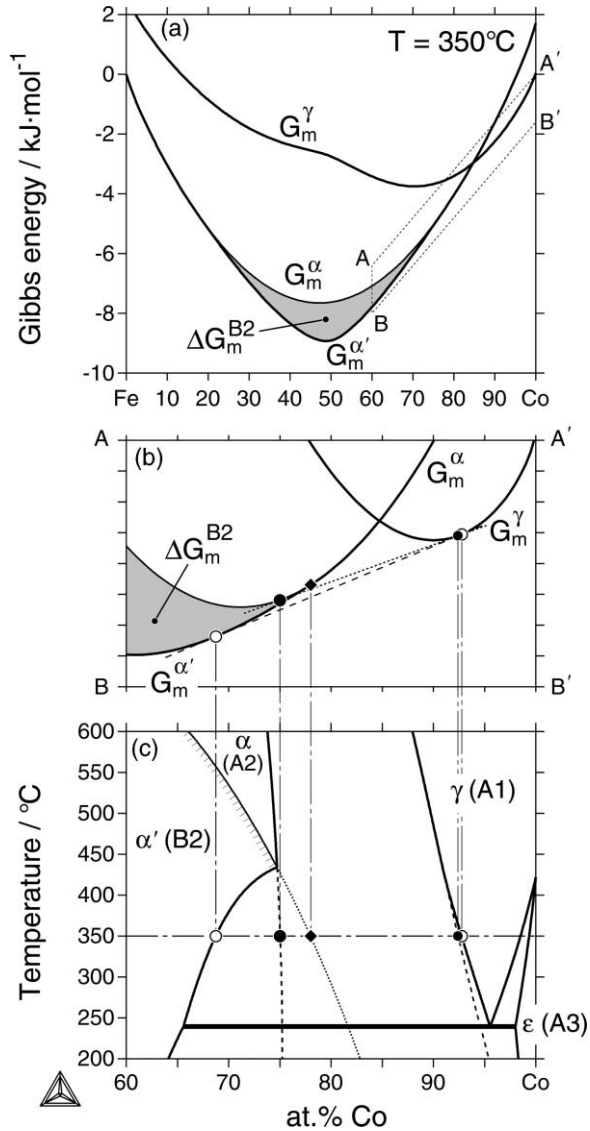

Fig. 13. Effect of B2 ordering on the phase equilibria between α (A2) and γ (A1): (a), (a′) calculated Gibbs energies at 350°C and (b) stable and metastable phase equilibria.

extends below the A2/B2 boundary. This characteristic feature is in good accordance with the experimental results of the present work as shown in Fig. 12(b).

5.3. Effect of B2 ordering energy on phase equilibria

Fig. 13 illustrates the effect of B2 ordering on the α (A2)/γ (A1) equilibrium. The ordering energy due to B2 configuration,

$$\Delta G_m^{B2} = G_m^{\alpha'(B2)} - G_m^{\alpha(A2)}, \tag{15}$$

is shown as a shaded area in Fig. 13(a) and (a′). Without the contribution of this excess ordering energy, the disordered b.c.c. phase α (A2) is in equilibrium with γ (A1), whose equilibrium compositions are indicated by solid circles in Fig. 13(a′) and (b). The shaded area in the partial diagram Fig. 13(a′) enlarged from a parallelogram AA′B′B in Fig. 13(a) shows the decrease in the

Gibbs energy of the b.c.c. phase due to the B2 ordering, which results in stabilization of the b.c.c. solution phase and extends the b.c.c. (α and α′) + f.c.c. (γ) two-phase region as shown by the open circles in Fig. 13(a′) and (b).

5.4. Effect of the short-range ordering (SRO)

The SRO contribution to the Gibbs energy is indispensable for calculating the phase equilibria of the order–disorder system. In the present calculation, this effect for the B2 ordering was approximated by introducing reciprocal parameters into the s-CEF model. Fig. 14 shows a comparison of (a) ordering energies and (b) A2/B2 boundaries calculated with and without taking the approximated SRO contribution into account. The ordering energies were described as follows:

$$\Delta G_m^{B2(SRO)} = G_m^{B2(SRO)} - G_m^{\alpha(A2)}, \quad (16)$$

$$\Delta G_m^{B2(LRO)} = G_m^{B2(LRO)} - G_m^{\alpha(A2)}, \quad (17)$$

and the net SRO contribution shown as a shaded area in Fig. 14(a) is calculated by

$$\Delta G_m^{SRO} = \Delta G_m^{B2(SRO)} - \Delta G_m^{B2(LRO)}. \quad (18)$$

The SRO contribution causes the stabilization of the disordered A2 phase around the A2/B2 boundary and results in a decrease in the A2/B2 transition temperature $T_o^{A2/B2(SRO)}$ compared with $T_o^{A2/B2(LRO)}$ as shown in Fig. 14(b). This result is in accordance with the theoretical calculation by Inden [45], which was calculated with the irregular tetrahedron cluster of b.c.c. crystal. Fig. 15 shows the calculated molar heat capacity under two conditions, namely, including and not including the SRO approximation. The calculated result with the SRO effect is in good agreement with the experimental data measured by Castanet and Ferrier [46].

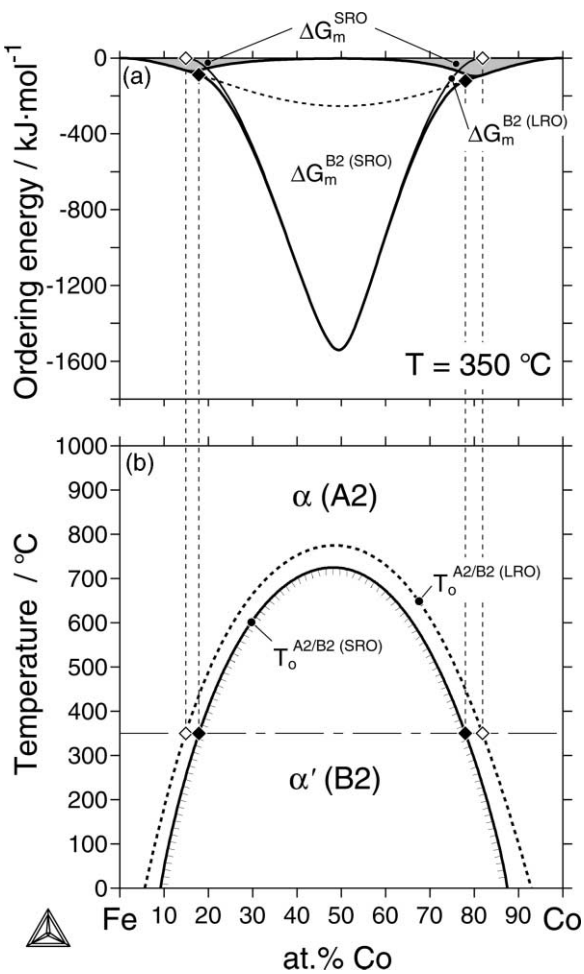

Fig. 14. Effect of the approximated contribution of SRO (a) on the B2 ordering contribution to the Gibbs energy at 350°C and (b) on the A2/B2 transition boundary.

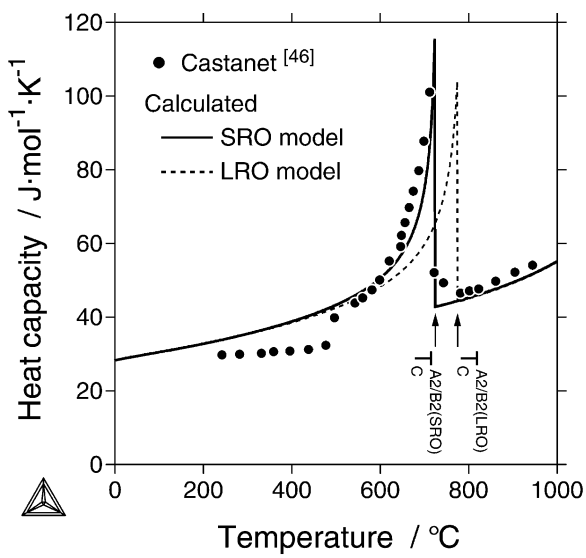

Fig. 15. Calculated molar heat capacity of the equiatomic Fe–Co alloy.

5.5. Effect of the interaction between chemical and magnetic ordering

It is well known that the magnetic properties, such as the Curie temperature T_c and the mean magnetic moment β, vary depending on the degree of ordering. For instance, T_c of the Ni_3Fe ($L1_2$) ordered compound is higher than that of the disordered solution (A1) phase [47]. On the other hand, both T_c and β of the Ni_3Pt ($L1_2$) and NiPt ($L1_0$) and T_c of the CoPt ($L1_0$) and $CoPt_3$ ($L1_2$) ordered compounds are strongly depressed due to the ordering compared with those of the metastable disordered Ni–Pt and Co–Pt alloys [30,48]. The Fe_3Al (DO_3) ordering also shows the same tendencies [49]. According to the B-W [37] and the CVM [10] calculations, the deviation of the A2/B2 boundary in the Fe–Co system is lower under the assumption of the fictitious non-magnetic state, which indicates that the interaction between the magnetic and chemical ordering increases the stability of the B2 ordered state. Experimental evidence of the enhanced stability of the ordered state due to the interaction was presented by Pierron-Bohnes et al. [50] for the short-range order parameters of the Fe–Co and Fe–Mn alloys. Ohnuma et al. [12] proposed that the enhanced stability due to the interaction between the chemical and magnetic ordering can be described by introducing an incremental deviation of T_c and β into the ordered phase using the conventional CALPHAD method, which were evaluated in the present calculation:

$$\Delta T_c^{B2/A2} = T_c^{B2}(\sigma = 1) - T_c^{A2}(\sigma = 0) = +370\ \mathrm{K}, \quad (19)$$

$$\Delta \beta^{B2/A2} = \beta^{B2}(\sigma = 1) - \beta^{A2}(\sigma = 0) = +0.10\mu_B, \quad (20)$$

where σ is the degree of B2 ordering. Fig. 16 shows the result of the calculation of (a) the Gibbs energy of the b.c.c. α (A2) and α′ (B2) phases at 200°C and (b) α (A2) / α′ (B2) transition boundary in a full interaction and a half interaction (i.e. $\Delta T_c^{B2/A2} = +185$ and $\Delta \beta^{B2/A2} = +0.05$) between the chemical and magnetic ordering. As shown in Fig. 16(a), the excess Gibbs energy of the B2 ordering ΔG_m^{B2} shown by the shaded area increases tremendously due to the enhanced ordering contribution caused by the interaction between the magnetic and chemical ordering, which results in an increase in the α (A2)/α′ (B2) transition boundary as illustrated in Fig. 16(b). Without the interaction (i.e. $\Delta T_c^{B2/A2} = \Delta \beta^{B2/A2} = 0$), the A2/B2 transition temperature of the equiatomic FeCo compound decreases further down to −80°C. This indicates that the enhanced stability due to the interaction between the chemical and magnetic ordering plays a key role in the formation of the B2 ordered phase.

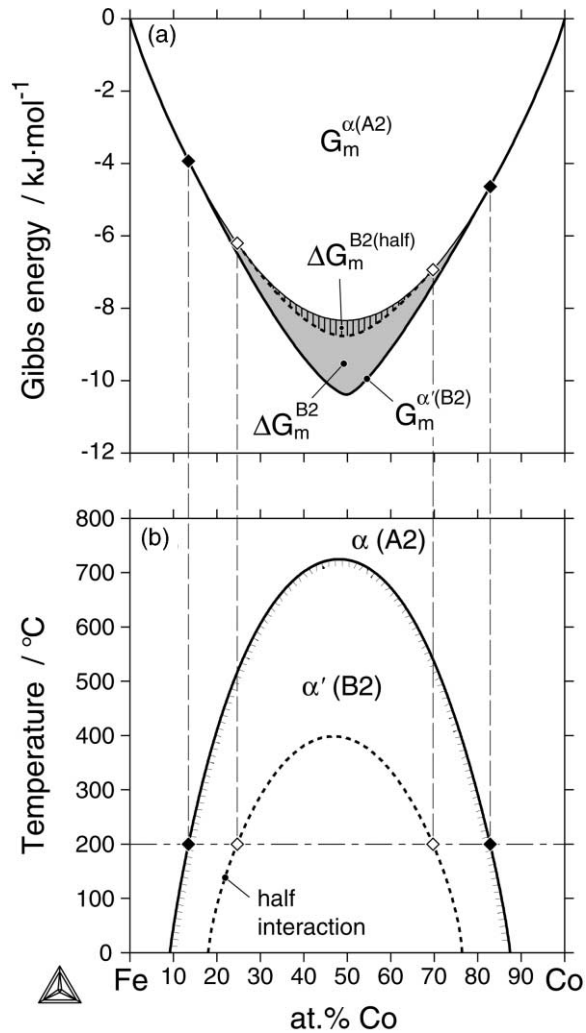

Fig. 16. Effect of the interaction between the magnetic and the chemical ordering (a) on the B2 ordering contributions to the Gibbs energy at 200°C and (b) on the α (A2)/α′ (B2) transition boundary.

6. Conclusion

Phase equilibria between the α (A2), α′ (B2) and γ (A1) phases in the Fe–Co binary system were determined using thin film specimens at temperatures between 400 and 800°C. It was found that the equilibrium state can be attained even at low temperature due to the enhanced diffusion through surface and grain boundary of thin-film specimens. The limiting temperature for obtaining the equilibrium was estimated to be about 0.4 T_m, where T_m represents the melting temperature. The α/α′/γ phase equilibria in the Fe–Co system were determined by means of lattice constant measurements of thin-film specimens. It was confirmed that the two-phase region between the b.c.c. and f.c.c. γ (A1) phases extends below the α (A2)/α′ (B2) order–disorder transition boundary.

Thermodynamic assessment of the Fe–Co binary system has been carried out on the basis of previous and present experimental data. The ordering contribution to the Gibbs energy of the α′ (B2) configuration and its interaction with the magnetic ordering were described under the split compound energy formalism (s-CEF). The short-range ordering (SRO) contribution to the Gibbs energy due to the pair probability was approximated by introducing the reciprocal parameters into the s-CEF description. The calculated phase diagram was found to be in good agreement with experimental data. The expansion of the two-phase equilibrium between the α′ and the γ phases at lower temperatures can be explained by the present calculation as being due to the effect of the B2 ordering.

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